Target Data Model v2 — Business-First Design

Oct 1, 2026 · @Rajat

The data model Ellementry's AI Hub should have, designed from the business and from what each agent must answer, with today's data mapped onto it and every gap made visible.

Design principles

The model is shaped by what Ellementry does and what each agent must answer; today's files are mapped onto it afterwards. Four rules hold throughout:

- **Business first.** Every table records a real business process (taking an order, moving stock, paying for ads), at the level of detail the decisions need, whatever today's exports happen to contain.
- **Gaps are part of the design.** Every table and field the business needs exists in the design, marked Have, Partial or Gap. A gap is listed with what would fill it. It is never filled with an estimate.
- **Context lives in the database.** Every business rule and file quirk that today only the person handling the file knows ("exports go through Dileep at cost", "CHANNELNAME is a location") is stored as a numbered rule. Every table, column and classification it affects points back to it, and the AI agents can read the rules too.
- **Built to extend.** All processes share one set of dimensions (date, product, location, channel and so on). Adding a source, a process, an attribute or an agent follows a fixed recipe, with no redesign.

Scope: this is the structured warehouse. Documents, images, review text and meeting notes go to the knowledge store (pgvector), linked to the same product and partner keys so agents can join the two.

What each agent must answer

Taken from the AI Hub document's agent responsibilities, alerts and weekly CEO report, made concrete with what we know about Ellementry. These questions decide which processes the model must record.

Agent	Questions it must answer	Build
Stock AI	Stock by SKU at every location, including stock held at franchise, SIS and quick-commerce partners. Days of cover, stockout and overstock risk. Demand forecast. What to reorder from which factory, and what to move from Bagru to which FBA or dark store. Ageing and discontinued stock	Now
Operations AI	Order-to-ship and ship-to-deliver time, SLA breaches, RTO rate by courier and pincode. Transfers in transit and how long. Return volumes and processing time. Supplier on-time and in-full delivery. Warehouse throughput	Now
Executive AI	Daily net sales (excl. GST) by channel, orders, average order value, blended ROAS, stock value and cover, top and bottom products, returns rate, all against last week and last year	Later
Marketing AI	Spend, ROAS, CTR, CPC and customer acquisition cost by campaign and platform; trends; where to move budget	Later
Commerce AI	Sales and conversion by marketplace and the website; listing health (live, rejected, suppressed); product performance; bottlenecks such as stockouts at FBA	Later
Product AI	Everything about one SKU: attributes, collection, price history, stock everywhere, sales, returns, lifecycle, images	Later
Finance AI	Gross margin by SKU and channel; channel contribution after marketplace fees, shipping and returns; EBITDA; cash flow; receivables from B2B partners and marketplace payouts	Later
Customer AI	Ratings, review themes and return reasons by product and channel; repeat customers	Later
Design AI	Sales and returns by collection, material and design; packaging damage; links to design assets	Later

Alerts (declining ROAS, low inventory, listing rejections, falling conversion, retailer issues) and the weekly CEO report read from the same tables, plus a small set of tables that store what agents produce (see Key changes).

Business processes

The business runs 23 processes the agents need, plus one for what the agents themselves produce. Today's files cover 13 of them fully or partly. 10 are built now for Stock and Operations AI; the rest are designed now and built when their source arrives.

Process	Table — one row per	Shared dimensions	Agents	Today's source	Coverage	Build
Sales and demand						
Customer orders	fact_order_line — order line, any channel	Date, Product, Channel, Location, Customer, Geography	Exec, Commerce, Stock	Order numbers on ERP invoices; WMS orders. No marketplace order reports (so no cancelled or unfulfilled orders)	Partial	Later
Company sales (book of record)	fact_sales_line — invoice line	Date, Product, Location, Channel, Partner, Geography, Entity	All	ERP Retail and Domestic Sales	Have	Now
Returns, RTO, credit notes	fact_return_line + receipts bridge — return line	Date, Product, Location, Channel, Partner, Return reason	Stock, Ops, Customer, Finance	ERP Retail and Domestic Returns; RTO status in WMS	Partial: no return reasons	Now
Partner sell-through	fact_partner_sellout — partner × day × store × SKU	Date, Product, Channel, Partner, Geography	Stock, Commerce	Blinkit and Instamart sales	Partial: no franchise or SIS	Now

Process	Table — one row per	Shared dimensions	Agents	Today's source	Coverage	Build
Stock at partners (consignment)	fact_consignment_position — partner × SKU × day	Date, Product, Partner, Location	Stock, Finance	Derived: sent to, returned by and sold through sale-or-return partners	Partial	Now
Prices and promotions	fact_price — SKU × channel × period	Date, Product, Channel	Commerce, Finance, Product	MRP in stock report and master; discounts on invoices	Partial: no promotion calendar	Later
Marketing						
Ad performance	fact_ad_performance — campaign × day	Date, Campaign, Channel, Product	Marketing, Exec	None (Google Ads, Meta Ads)	Gap	Later
Website and marketplace traffic	fact_traffic — channel × day (× product)	Date, Channel, Product	Commerce, Marketing	None (GA4, Shopify, seller panels)	Gap	Later
Listing health	fact_listing_status — listing × day	Date, Product, Channel	Commerce	None (seller panels)	Gap	Later
Supply chain						
Purchasing	fact_purchase_order_line — PO line, latest state	Date, Product, Supplier	Stock, Ops, Finance	ERP Pending PO report	Partial: export date, factory list	Now
Goods receipt	fact_goods_receipt_line — receipt line	Date, Product, Location, Supplier	Stock, Ops	GRN fields in the PO report (one receipt per line)	Partial	Now
Production at Dileep factories	fact_production_order — production order line	Date, Product, Supplier	Stock, Ops	None	Gap	Later
Stock transfers	fact_stock_transfer_line — transfer line	Date, Product, Location (from, to)	Stock, Ops	ERP Stock Transfer out and in	Have	Now
Stock movements (ledger)	fact_stock_movement — movement × location × SKU	Date, Product, Location, Movement type	Stock, Ops, Finance	Derived from sales, returns, receipts, transfers	Partial: no adjustments, write-offs or production receipts	Now
Stock position	fact_inventory_snapshot — day × location × SKU	Date, Product, Location	Stock, Exec, Ops	ERP stock report; Blinkit, Instamart inventory	Partial: history starts now; no Amazon FBA report	Now

Process	Table — one row per	Shared dimensions	Agents	Today's source	Coverage	Build
Order fulfilment	fact_shipment + SKU bridge — outbound delivery	Date, Channel, Location, Courier, Geography	Ops, Customer	WMS dispatch	Partial: file not yet received	Now
Courier cost	fact_courier_charge — shipment	Date, Courier, Geography	Ops, Finance	None	Gap	Later
Finance						
Marketplace settlements and fees	fact_settlement_line — order × fee type	Date, Channel, Product	Finance, Commerce	None	Gap	Later
Ledger and P&L	fact_gl_entry — ledger entry	Date, Account, Entity, Location	Finance, Exec	None	Gap	Later
Product cost	fact_product_cost — SKU × effective date	Date, Product, Supplier	Finance, Stock	Cost Rate on some invoices; PO purchase price	Partial	Later
Receivables and payouts	fact_receivable — invoice × due date	Date, Partner, Channel	Finance	None	Gap	Later
Customer and team						
Reviews and ratings	fact_review (text in knowledge store) — review	Date, Product, Channel, Customer	Customer, Design, Commerce	None	Gap	Later
Team productivity	to define with HR	Date, Employee	Exec	None	Gap	Later
Agent outputs						
Forecasts, recommendations, alerts	insight tables — one output each	Date, Product, Location, Channel	All	Written by the agents	n/a	Now (Stock AI)

Shared dimensions

Fifteen dimensions are shared by every process, so a number from one agent always lines up with another's. Product is the most important one, and also the biggest gap.

Dimension	What it describes	Ideal attributes	Today	Gaps
Date	Calendar day	Calendar, Indian fiscal year, ISO week; festive and sale-event flags (Diwali, marketplace sale days)	Have (generated)	Festive and sale-event calendar
Product	Every item code, with history	SKU, name, item type, collection, category, material, room, colour, size, weight, pack ("Set of 2"), HSN, MRP, cost, launch and discontinue dates, set components, EAN, image link	SKU, name, type, unit, status (Continued / Discontinued), HSN, partial category	Collection, material, room, launch and discontinue dates, set composition, dimensions and weight, images
Location	Every place stock sits	ERP name, type, operator, city, state, sellable, stock system of record, active, opening and closing dates	Have: 74 locations mapped, under review	Opening and closing dates; type of some stores (own, SIS or franchise)
Channel	Where a sale happens	Channel, sub-channel, group, fulfilment model (FBA, self-ship, Seller Flex), sale-or-return, commission rate	Have: 22 channels	Commission and fee schedule

Dimension	What it describes	Ideal attributes	Today	Gaps
Partner	Business counterparty	Name, role (customer or supplier), type, group, sale-or-return, related party, GSTIN, credit terms	Types and rules defined	Franchise, SIS and quick-commerce names; credit terms
Supplier	Factory goods come from (a partner role)	Factory, group company or external, standard lead time, location	Suppliers on POs	Factory list
Customer	A D2C buyer, anonymised	Customer key (from a one-way hash of phone or email), first order date, pincode, acquiring channel	Data in raw only (personal data)	A decision on hashing; marketplaces mask buyers ("Amazon Customer"), so mainly website buyers can be identified
Geography	Delivery place	Pincode, city, district, state, region, courier serviceability	Pincodes on invoices and dispatch	Pincode master (public India Post data)
Courier	Who delivers	Courier, service level (surface, express), aggregator (Shiprocket, Delhivery)	Names in WMS and ERP	Standardised list

Dimension	What it describes	Ideal attributes	Today	Gaps
Return reason	Why it came back	Reason, group (damaged, not as described, changed mind, undeliverable), who is at fault	None	ERP returns carry no reason; marketplace return reports do
Movement type	Why stock moved	Sale, return, transfer out / in, receipt, production, adjustment, write-off	Defined	None
Campaign	An ad campaign	Platform, account, campaign, ad set, objective	None	Ads data
Entity	Legal entity and GST registration	Dileep Essentials Pvt Ltd per state GSTIN; group companies	GST state codes	GSTIN list
Account	Ledger account	Chart of accounts, P&L grouping	None	Ledger export
Employee	Staff member	Name, team; used for exhibition stock and team productivity	Names on domestic invoices	HR list (later)

Key changes from the v1.3 model

The raw layer, file contracts, loaders, reference mappings and core conventions stay as they are; the core tables are reshaped. They hold no data yet, so they are replaced cleanly, with no data migration.

Change	Why	What it gives Stock and Operations AI
Stock ledger as the backbone. Every stock movement (sale, return, transfer, receipt, production, adjustment) becomes one signed row per location in fact_stock_movement	Stock questions are about flows, not only levels	Yesterday's stock plus today's movements must equal today's snapshot; any difference is reported as unexplained, which exposes missing movement types. History before the first snapshot can be rebuilt backwards
Stock at partners is still stock. Goods sent to franchise, SIS and quick-commerce partners stay Ellementry's stock at a partner location until they sell through or come back	Business-confirmed sale-or-return	True stock and cover per SKU, not understated by stock sitting at partners
Order, invoice, shipment and return are separate, linked documents	One order can have several invoices, shipments or returns	Order-to-ship, RTO and return rates computed per order, not per invoice
Returns keep both dates. The credit-note date and each goods-receipt date are stored; a bridge holds the several receipts one credit note can cover	Found by the loader: one credit note covered receipts months apart	Net sales and stock timing each use the right date
Goods receipts become their own table, separate from PO lines	Partial deliveries	Supplier lead time and in-full delivery per receipt
Product carries full history and set composition	MRP and status change; "Set of 2" items deplete component stock	Correct stock for sets; price and lifecycle as at any date
Agent outputs are stored in a new insight schema: forecasts (with model version), recommendations (with the human decision taken), alerts	An agent's advice must be auditable and its accuracy measurable	Forecast accuracy tracking; a record of which reorders were accepted

Change	Why	What it gives Stock and Operations AI
Context register in the database (next section)	Business rules and file quirks must not live only in people's heads	Every number traceable to the rule behind it

Making buried context visible

Every fact about the data that someone explained to us, or that we found in a file, becomes a row in the database that anyone, and any agent, can query. Five registers hold it:

Register	Holds	Example
meta.business_rule	Each business rule: a numbered statement, what it applies to, who confirmed it and when, status, and what replaced it if superseded	BR-007: Exports are routed through Dileep group entities and billed cost-to-cost. Applies to Domestic Sales partner classification. Confirmed by the business, Sep 2026
meta.source_note	What only the person handling a file knows: column meanings, traps, how and when it is exported, by whom	ERP Retail Sales: PRODASSESSABLEVALUE includes GST; CHANNELNAME is the billing location, not the channel
meta.column_dictionary	Every warehouse column: business definition, the source file and header it comes from, how it is derived, and the rules behind it	fact_sales_line.net_sales_amount = Retail PRODDOCVALUE or Domestic Taxable; rules BR-001, BR-002

Register	Holds	Example
meta.data_gap	Every gap in this design: the missing table or field, what would fill it, owner, target phase	Return reason: none in ERP returns; marketplace return reports carry it
meta.source_coverage	Which source covers which period, per month: full, partial or none	Blinkit sales: from Sep 2026 only; franchise sell-through: none

How one buried fact travels through the model, for example "exports go through Dileep at cost":

1. It is recorded as rule BR-007, with who confirmed it and when.
2. The partner mapping marks the Dileep accounts as type GROUP_EXPORT, citing BR-007.
3. Every Dileep invoice reaches the dashboards as channel "Export via Dileep" through that partner row, so the rule applies to all history at once.
4. The data dictionary lists BR-007 against the channel and margin columns.
5. When an agent is asked why export margin is near zero, it can read BR-007 and say so.

Gaps work the same way. A dashboard or agent never shows a gap as zero: it reads meta.source_coverage and says, for example, "franchise sell-through: no data yet". Agents are also given the rules and the dictionary as part of the knowledge store, so their answers follow the same definitions as the dashboards. The issue log moves into the database as meta.data_issue once it settles; until then the Excel log stays the working copy.

Adding to the model later

Each kind of addition follows one fixed recipe, so the model grows without redesign and without anyone remembering how it was done last time.

To add	Recipe	What stays untouched
A new source (e.g. Amazon's FBA inventory report)	Register it with its source notes; generate its contract from a sample file; generate its raw table; write its staging step; map its values (the coverage check must show nothing missing); point it at the existing table for its process; update coverage and close its gap	Every existing table and loader
A new process (e.g. ad performance)	Add one fact table built on the existing shared dimensions; add a dimension only if it is truly new (campaign); add its row to the process table; add mart views	Every existing fact
A new attribute (e.g. product collection)	Add the column to its dimension, declaring whether its history is kept; map its source; close its gap	Every fact
A new location, channel or partner	Add a row to the reviewed reference file	All code
A new business rule	Record it in the rule register first; the change that implements it cites the rule number	Rules it does not supersede
A new agent	Map its questions to processes in the table above; it reads marts and the knowledge store and writes to the insight schema	The warehouse, unless it needs a new process

Three safeguards apply to every change: numbered migration files (never edits to applied ones) with the schema version recorded in the database; all tests passing before a change is used; and the coverage check run on every data load. Moving to the cloud keeps the same SQL and contracts. Large tables are split by month, and loading is scheduled instead of run by hand.

Gap register

24 gaps, sorted by when they block something. The first 6 limit the Stock and Operations AI prototype; the last 8 only matter for later agents. All of them will also live in meta.data_gap.

Priority	Gap	Needed by	How to fill it	Owner
High	Daily stock history	Stock	Save the ERP, Blinkit and Instamart stock reports every day from now	Business, ERP team
High	Stock adjustments, damage write-offs, production receipts into Bagru	Stock (ledger must balance)	ERP stock journal or adjustment register export	ERP team
High	Full WMS dispatch export	Operations	Export for the whole period, as CSV	Business
High	Billing names of franchise, SIS and quick-commerce partners	Stock, Finance	List from the business	Business
High	Factory list	Stock (lead times)	List from the business	Business
High	Which date is the domestic return date	Stock, Finance	Decision	Business

Priority	Gap	Needed by	How to fill it	Owner
Medium	Amazon FBA inventory and inbound shipment reports	Stock (FBA stock; transfers never received)	Seller Central exports	Business
Medium	Franchise and SIS sell-out	Stock	Partners' sales reports	Business
Medium	Return reasons	Operations, Customer	Marketplace and Shopify return reports	Business
Medium	Set composition ("Set of 2" and similar)	Stock	Product team list, or bill of materials from the ERP	Business
Medium	Product hierarchy: collection, material, room	Stock, Design, Product	Product team; the SKU code pattern may supply part of it	Business
Medium	Courier cost per shipment	Operations, Finance	Courier and aggregator invoices (Delhivery, Shiprocket)	Business
Medium	Location opening and closing dates; store types	Stock	Review of the location mapping workbook	Business
Medium	Marketplace order reports, incl. cancellations	Stock (true demand), Commerce	Seller panel exports	Business
Low	Pincode master	Operations	Public India Post data	Data team
Low	Festive and sale-event calendar	Stock (forecasting)	List from the business	Business
Later	Ad spend and performance	Marketing, Executive	Google Ads and Meta Ads, via connectors	Data team

Priority	Gap	Needed by	How to fill it	Owner
Later	Website and marketplace traffic; listing status	Commerce	GA4, Shopify, seller panels	Data team
Later	Marketplace settlements and fees	Finance	Settlement reports from each marketplace	Business
Later	Ledger, P&L, receivables	Finance, Executive	ERP accounts export	ERP team
Later	Product cost (standard or landed)	Finance	Costing sheet or ERP	Business
Later	Reviews and ratings	Customer, Design	Marketplace and Shopify reviews	Data team
Later	Customer identity policy	Customer, Executive	Decision on hashing phone and email (India's DPDP Act applies)	Business
Later	Team data	Executive	HR system	Business

Decisions and next steps

Two decisions shape the prototype tables; the other two can wait.

Decision	Proposal	Needed
Stock at sale-or-return partners	Counted as Ellementry's stock at the partner until it sells through or comes back	Before the build
Domestic return date	Credit-note date for net sales; goods-receipt dates for stock timing	Before the build
Storing agent outputs	Yes: every forecast, recommendation and alert, with the human decision taken	Default unless you object

Decision	Proposal	Needed
Customer identity	One-way hash of phone or email, no personal data in the warehouse	Before Customer AI

Next steps:

1. You review this design and correct anything that doesn't match how the business works.
2. I write the v2 core tables and the five context registers, seeded from the Business Rules Register and the issue log, with tests.
3. Staging and core build for the 10 processes marked Now.
4. Mart views and dashboards for Stock and Operations AI.

Build status

Schema v2.0 implements this design: the five context registers, the v2 core tables and the insight schema, with 76 schema tests, 6 seed-loader tests and 49 loader tests passing. Where the build refined the design:

- The two "before the build" decisions were taken as approved with this document and recorded as rules BR-022 and BR-023 (confirmation: design approval).
- Stock at partners is read from the stock ledger at partner locations, so it needs no table of its own.
- Four dimensions were added: order, courier, return reason and movement type (16 movement types).
- Two virtual locations were added: "With employees (exhibitions)" and "Blinkit (Kiranakart) consignment stock".
- The gap register holds 26 gaps: the receiving location on goods receipts and the location column on transfer exports were added.
- A safety switch keeps unlisted partner names unclassified, rather than classified by elimination, until the business confirms the partner list is complete.